RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM – 602 105

RAJALAKSHMI
ENGINEERING
COLLEGE

CS23334 – Fundamentals of Data Science Lab

Laboratory Record Note Book

Name : M. MAADHAVAN

Year / Branch / Section : II - CSE - H

University Register No. : 2116250701394

College Roll No. : 250701394

Semester : III

Academic Year : 2025-2029

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23334 – Fundamentals of Data Science Lab

NAME	M.MAADHAVAN
ROLL NO	250701394
YEAR	II
SEC	H

RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution

BONAFIDE CERTIFICATE

Name: M.MAADHAVAN

Academic Year: 2026-2027 **Semester:** III **Branch:** CSE

Register No. 250701394

Certified that this is the bonafide record of work done by the above student in the FUNDAMENTALS OF DATASCIENCE ***Laboratory during the academic year 2026 – 2027.***

3/11/26

Signature of Faculty in-charge

Submitted for the Practical Examination held on 05/11/26

Internal Examiner

External Examiner

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Sl.NO	Experiment	Date	GitHub
1.a	Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.	29/07/26	
1.b	Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.	29/07/26	
1.c	Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.	05/08/26	
1.d	Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.	05/08/26	
2	Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.	12/08/26	
3.a	Conduct an experiment to show data visualization using line plot	19/08/26	
3.b	Conduct an experiment to show data visualization using bar chart	19/08/26	
3.c	Conduct an experiment to show data visualization using pie chart	19/08/26	
4	To Count the frequency of occurrence of a word in a body of text is often needed during text processing.	26/08/26	
5	Data Collection and Initial Exploration	16/09/26	
6	Time Series Analysis of Website Traffic	16/09/26	

7	Random Sampling and Sampling Distribution	23/09/26	
8	Conduct Z test for the Data given using Python	23/09/26	
9	Experiment to understand Matplotlib library use cases in Data Science through visualization.	23/09/26	
10	Experiment to understand array function in Data science.	23/09/26	
11	Experiment to understand pandas library use cases in Data science.	30/09/26	
12	Experiment to detect outliers in a given data set.	30/09/26	
13	Experiment to handle missing values and inappropriate data in a given data set.	30/09/26	
14	Experiment to understand feature scaling	30/09/26	
15	Experiment to understand the data preprocessing in Data science	30/09/26	
16	Experiment to understand EDA-Quantitative and Qualitative analysis.	07/10/26	
17	Experiment to understand Linear Regression for a given data set.	07/10/26	
18	Experiment to understand KNN algorithm for a given data set.	07/10/26	
19	Experiment to understand Logistic Regression for a given data set.	07/10/26	
20	Experiment to understand K-means clustering algorithm for a given data set.	07/10/26	